

University of Hohenheim

**Time Series Approaches to Cocoa Price
Risk**

Master Thesis

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1 Introduction

Volatility forecasting is fundamental to financial risk management, portfolio allocation, and derivative pricing (Poon and Granger 2003). Two modeling paradigms dominate the literature. The first treats volatility as latent and inferred from returns; the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model of Bollerslev (1986) is the canonical approach. The second, enabled by high-frequency data, treats volatility as directly observable through realized measures, and the Heterogeneous Autoregressive (HAR) model of Corsi (2009) has become the dominant forecasting framework within this paradigm, capturing long-memory properties of volatility in a parsimonious, OLS-estimable structure.

The majority of volatility forecasting research focuses on conditional mean forecasts: the expected level of future volatility given current information. Yet the conditional mean provides an incomplete characterization of future volatility. When the volatility distribution is skewed or fat-tailed, as is typical for commodity markets, the conditional quantiles can diverge substantially from the mean, and understanding how the entire distribution shifts with predictor values becomes important. Quantile regression, introduced by Koenker and Bassett (1978), provides a natural framework for estimating conditional quantiles directly, without distributional assumptions.

Cocoa futures present a particularly relevant case for distributional volatility forecasting. Production is geographically concentrated in West Africa, biologically slow to adjust, and exposed to both disease and climate shocks. Between 2022 and 2024, international cocoa prices approximately tripled, an increase unprecedented in the commodity's modern trading history. Despite this salience, dedicated cocoa volatility research remains limited, and the application of HAR-based quantile regression to cocoa, or to any agricultural commodity, is an open gap in the literature. At long forecast horizons, where the autoregressive signal from lagged volatility fades, climate variables offer an additional source of predictive information. The El Niño–Southern Oscillation (ENSO) influences West African rainfall patterns with effects on cocoa yields hypothesised to materialise 6–12 months after the climate signal. Evidence from oil and broad agricultural commodity surveys suggests that ENSO can improve volatility forecasts at these longer horizons (Bouri et al. 2021; Bonato et al. 2023), motivating a test of whether this channel extends to cocoa.

This thesis addresses two research questions:

RQ1 Can HAR-based quantile regression models forecast cocoa futures volatility better than baseline models (historical volatility, GARCH, HAR-OLS) at horizons from 1 day to 12 months?

RQ2 Does the ENSO climate index improve volatility forecasts, especially at long horizons?

We address these questions by estimating HAR-QR models at multiple quantiles ($\tau \in \{0.50, 0.75, 0.95\}$) and horizons (1 day, 1 week, 1 month, 3 months, 6 months, 12 months). At each horizon, we compare out-of-sample forecast accuracy against standard benchmarks: historical rolling volatility, GARCH(1,1), and HAR estimated by ordinary least squares. For long horizons, we test whether adding ENSO improves forecasts, with particular focus on the upper quantile ($\tau = 0.95$).

The thesis makes three contributions. First, it provides the first application of HAR-QR to an agricultural commodity, extending a framework previously confined to equity and precious-metals markets. Second, it tests HAR-QR at horizons of 6–12 months, far beyond the current literature limit of approximately one month (Lyócsa and Molnár 2021). Third, it tests whether ENSO enhances quantile volatility forecasts for cocoa, a combination untested in the literature.

The remainder of the thesis is organized as follows. Section 2 reviews the literature on volatility measurement, HAR models, quantile regression in finance, and climate-commodity linkages, establishing the research gap this thesis addresses. Section 3 specifies the HAR-QR model, the multi-horizon projection approach, and the evaluation framework. Section 4 describes the ICE London cocoa futures data and validates the daily volatility proxy against realized volatility from intraday data. Section 5 presents empirical results organized by forecast horizon. Section ?? interprets findings, discusses limitations, and considers practical implications.

2 Literature Review

This section reviews the literature underlying volatility forecasting for cocoa futures. We begin with the cocoa market itself, then trace how volatility measurement evolved from simple estimators to the HAR framework. We then review quantile regression for volatility and ENSO as a climate predictor at long horizons, followed by the evaluation methods used to compare forecasts. The intersection of these strands identifies the gap this thesis addresses: no study applies HAR-based quantile regression to any agricultural commodity, and no HAR-QR study extends beyond approximately one month.

2.1 The Cocoa Market

Cocoa occupies an unusual position among agricultural commodities. Its production is geographically concentrated, biologically slow to adjust, and structurally exposed to climate variability in ways that set it apart from grains or oilseeds. These features interact to produce a market where supply disruptions are both frequent and slow to resolve, precisely the conditions under which distributional, rather than mean, volatility

forecasts become most valuable.

The West African cocoa belt accounts for approximately 70% of world production, with Côte d’Ivoire and Ghana alone responsible for more than half of global output (Schroth et al. 2016; Läderach et al. 2013). Unlike grains or oilseeds, whose production is spread across multiple continents, cocoa’s concentration means that disruptions in a single region can move global prices substantially. Production is overwhelmingly smallholder-based, with approximately 90% grown on farms smaller than five hectares (Bensch et al. 2022). Since the mid-2000s, growing financial participation has added a further source of volatility (Bensch et al. 2022; Gilbert and Morgan 2010; Rogna and Tillie 2025).

A critical structural feature is the cocoa tree’s biological rigidity. New trees require five to seven years before reaching full productive capacity, with an economic lifespan of 30 to 40 years (Asante et al. 2025). Unlike annual crops, cocoa supply responds to market signals slowly, and the resulting low short-run supply elasticity amplifies price volatility whenever demand or supply shocks occur.

Cocoa is also among the most climate-sensitive perennial crops. West Africa is characterized by variable climates (Läderach et al. 2013), and drought during the critical September–December pod development period reduces yields substantially. The El Niño–Southern Oscillation (ENSO) is the primary near-term mechanism through which climate variability affects cocoa supply; Section 2.5 discusses this channel and its implications for long-horizon volatility forecasting.

The most striking recent illustration of these compounding dynamics is the 2022–2024 price crisis, during which international cocoa prices approximately tripled (Rogna and Tillie 2025). The crisis was driven by the convergence of the Cocoa Swollen Shoot Virus Disease (CSSVD) in Ghana and adverse El Niño–related drought across Côte d’Ivoire.

Despite the salience of cocoa in global commodity markets, the volatility forecasting literature has paid it limited attention. Bensch et al. (2022) provide the most comprehensive study of cocoa price determinants, finding that fundamentals are quickly integrated into prices, but do not extend to multi-horizon forecasting or tail-risk quantification. Bergander (2025) apply LSTM neural networks with weather data but find no improvement in point forecasts, plausibly because markets anticipate weather shocks before physical effects materialize. Within broader commodity surveys, Alfeus and Nikitopoulos (2022) include cocoa among 22 markets and find that monthly volatility is the most informative component, evidence of slow-moving structural dynamics, but offer no focused analysis. No study applies HAR-based models specifically to cocoa volatility, and none has examined quantile-based tail-risk forecasting for this market.

2.2 The Evolution of Volatility Measurement

A foundational challenge in volatility research is that volatility is latent: unlike prices, which are directly observable, it must be estimated from price movements (Poon and Granger 2003). Every volatility measure is an approximation, and the choice of estimator shapes both forecast quality and evaluation validity (Patton 2011). The measurement literature has evolved through three successive approaches, each reducing the statistical noise of the volatility proxy.

The earliest volatility estimates used only closing prices. The random walk model, exponential smoothing, and rolling-window standard deviations all extrapolate past variation into the future (Poon and Granger 2003). These methods are simple and require minimal data, but the squared close-to-close return is a highly imprecise estimator of daily variance. Nonetheless, Poon and Granger (2003) review 93 studies and find that “for forecast horizons that are longer than 6 months, a simple historical method using low frequency data over a period at least as long as the forecast horizon works best,” a finding that motivates the inclusion of rolling historical volatility as a benchmark in this thesis.

A significant efficiency improvement came from exploiting the full daily price range. Parkinson (1980) demonstrated that the high-low range achieves efficiency gains of approximately five times over close-to-close methods. Rogers and Satchell (1991) extended this to allow for non-zero drift. However, both estimators assume continuous trading, an assumption violated by commodity futures markets, which experience regular overnight closures.

Yang and Zhang (2000) synthesized these approaches into an estimator that is unbiased, drift-independent, and robust to opening jumps. By estimating overnight and trading-session variance separately and combining them optimally, the Yang-Zhang estimator achieves efficiency gains of 7–8 times over close-to-close methods. For commodity markets, where overnight information frequently causes opening gaps, this robustness is particularly valuable.

The availability of intraday transaction data in the late 1990s enabled a fundamentally different approach. Andersen and Bollerslev (1998) and Andersen et al. (2001) demonstrated that realized volatility (the sum of squared intraday returns) provides a far more precise measure of daily variance, resolving much of the apparent poor performance of earlier models. However, high-frequency data are plagued by microstructure noise and are often unavailable for commodity markets over long samples. Clements et al. (2024) provide evidence that HAR models applied to daily OHLC-based volatility achieve comparable forecasting performance at medium and long horizons, validating the use of daily data when intraday records are limited, as is the case for cocoa futures over multi-decade samples.

These developments establish the Yang-Zhang estimator applied to daily OHLC data as the volatility proxy for this thesis. The following section turns from the question of what to measure to the question of how to model and forecast its temporal dynamics.

2.3 The Heterogeneous Autoregressive Model

Two modeling frameworks are central to this thesis: GARCH, which serves as the main benchmark, and the Heterogeneous Autoregressive (HAR) model, which is the primary forecasting framework. Both capture a defining feature of volatility that range-based estimators ignore: its tendency to cluster over time.

Volatility clustering was first formalized by Engle (1982), and Bollerslev (1986) generalized this into the GARCH framework, capturing both clustering and mean reversion. The GARCH(1,1) has proven remarkably difficult to beat: Hansen and Lunde (2005) evaluate 330 volatility models and conclude that “nothing beats a GARCH(1,1),” establishing it as the natural benchmark. However, GARCH cannot replicate the slow, hyperbolic decay of volatility autocorrelations observed empirically, a property known as long memory. As Corsi (2009) observes, “standard GARCH and stochastic volatility short-memory models appear like white noise once aggregated over longer time periods.”

The HAR model, introduced by Corsi (2009), addresses this limitation through a parsimonious structure that captures long memory without the estimation difficulties of fractionally integrated models. It builds on the *Heterogeneous Market Hypothesis* of Müller et al. (1997), who documented that coarse-grained volatility (e.g., weekly) predicts fine-grained volatility (e.g., daily) more strongly than the reverse. This asymmetry reflects the presence of traders with different investment horizons: intraday speculators, swing traders, and long-term institutional investors, whose actions generate volatility that cascades across frequencies. Corsi formalized this insight into an additive model with daily, weekly, and monthly volatility components: a restricted AR(22) estimable by OLS that reproduces the autocorrelation patterns of long-memory processes.

Since its introduction, the HAR framework has been extended with jump components, realized semivariance, and time-varying coefficients (Andersen et al. 2007; Patton and Sheppard 2015; Tian et al. 2017). However, for agricultural commodities, Degiannakis et al. (2022) find that “sophisticated HAR-type models are not capable of outperforming the simple HAR in an out-of-sample exercise.” This sets a high bar for any augmentation to clear, and motivates subjecting any extension to rigorous out-of-sample testing before drawing conclusions.

For cocoa specifically, Alfeus and Nikitopoulos (2022) include it among 22 commodities in a broad HAR study, finding that “monthly volatility matters the most” for

cocoa, evidence that longer-term participants exert particular influence on commodity volatility dynamics. Haugom et al. (2016) provide complementary evidence that “the effect from the monthly volatility component is strongest for all assets when predicting conditional tails.” Despite this, no study focuses specifically on cocoa volatility dynamics or applies HAR to horizons beyond one month.

2.4 Quantile Regression for Volatility

Standard volatility models, whether GARCH or HAR, estimate the conditional mean of volatility. Yet when the underlying process exhibits fat tails and regime shifts, the conditional quantiles can diverge substantially from the mean. Quantile regression, introduced by Koenker and Bassett (1978), estimates conditional quantiles directly, with each quantile having its own parameter vector. This allows the relationship between predictors and the dependent variable to differ across the distribution without imposing parametric assumptions on the error distribution.

Baur and Dimpfl (2019) demonstrate that quantile autoregression encompasses both the conditional mean and conditional variance of the return distribution through the inter-quantile range. Their approach is robust to jumps and produces variance forecasts comparable to, and in some cases more accurate than, GARCH and HAR alternatives. Haugom et al. (2016) introduced HAR-QREG, combining the multi-horizon HAR structure with quantile regression, and show that the model passes conditional coverage tests at substantially higher rates than historical simulation.

HAR-QR has since been applied in equity and precious-metals markets. Lyócsa and Molnár (2021) develop a complete subset quantile regression variant achieving significant improvements at horizons up to 22 trading days; Huang et al. (2022) confirm the model’s superiority using high-frequency data for Chinese indices; and Song et al. (2025) show that HAR-QR parameters improve machine learning forecasts for gold.

Despite this growing body of work, two gaps remain. First, no study applies HAR-QR to any agricultural commodity (Degiannakis et al. 2022; Bonato et al. 2023). Second, existing studies are confined to short horizons: Lyócsa and Molnár (2021) extend to 22 trading days, the longest in the literature. Whether quantile forecasts retain predictive value at horizons of 6–12 months remains untested.

2.5 Climate Variables and Commodity Volatility

At long forecast horizons, the autoregressive signal from lagged volatility fades. Climate variables offer a potential source of predictive information that operates on precisely these longer timescales.

The El Niño–Southern Oscillation (ENSO) is an irregularly periodic fluctuation in Pacific sea surface temperatures that affects weather patterns globally, cycling between

warming (El Niño) and cooling (La Niña) phases every two to seven years. For cocoa, the transmission mechanism is direct: during El Niño phases, West Africa typically receives below-normal rainfall during the growing season, reducing yields in subsequent harvests. Schroth et al. (2016) document that “drought years regularly affect cocoa yields in West Africa and have particularly done so during severe El Niño years.” With the main crop harvested from October through March and the mid-crop from May through August, ENSO-driven rainfall anomalies are hypothesised to translate into production changes six to twelve months after the climate signal, a transmission window that, if confirmed empirically, would align with the long forecast horizons this thesis examines. Importantly, markets respond not only to realized weather but to climate expectations: Bergander (2025) observe that roughly 85% of the eventual price impact of a drought shock is reflected in futures prices before the actual crop loss occurs. This forward-looking price adjustment does not preclude predictability in *volatility*: even when the expected future price level rapidly impounds climate information, the conditional dispersion of realized outcomes may remain systematically forecastable from the ENSO state.

At the commodity price level, Ubilava (2018) examines ENSO–price relationships across 43 commodities, finding significant effects for tropically grown agricultural products including cocoa, with “more amplified price responses during El Niño events.” Recent work has extended this analysis from price levels to volatility. Bouri et al. (2021) find that ENSO improves oil volatility forecasts at horizons of 2–4 years, far beyond where lagged volatility alone provides value. Bonato et al. (2023) extend this to 16 agricultural commodities, finding that “the evidence of predictive value of El Niño and La Niña events strengthens at the longer term forecast horizons,” precisely where standard autoregressive models lose power. More broadly, Guo et al. (2025) demonstrate that climate risk indices significantly improve commodity volatility forecasts, with models incorporating climate risk “outperforming traditional ones in economic value.”

These findings establish ENSO as a physically motivated, exogenous predictor with demonstrated value for oil and agricultural commodity volatility at long horizons, but whether this extends to cocoa remains untested, and no prior study has combined ENSO with quantile regression.

2.6 Evaluating Volatility Forecasts

A distinctive challenge in evaluating volatility forecasts is that the variable of interest is unobservable, even *ex post* (Patton 2011). Every volatility measure is an imperfect proxy, and noise in the proxy can produce misleading model rankings depending on the loss function used.

Patton (2011) derives conditions for loss functions that yield rankings invariant to

the choice of volatility proxy. Among common loss functions, only mean squared error (MSE) and QLIKE satisfy these conditions. Non-robust loss functions can produce rankings that deviate by factors of one quarter to three, explaining widespread conflicting findings across studies. Between the two robust measures, QLIKE penalizes under-prediction proportionally more heavily, making it suitable when the forecasting goal involves risk assessment (Lyócsa and Molnár 2021; Clements et al. 2024; Ghysels and Marcellino 2018). Mean absolute error (MAE) is a further robustness check that captures deviations in the volatility level rather than its square.

Hansen et al. (2011) propose the Model Confidence Set (MCS) for simultaneous comparison of multiple models, identifying the set $\mathcal{M}^*$ containing the best model at a given confidence level. The MCS “acknowledges the limitations of the data”: informative samples yield small confidence sets, while uninformative samples include many models, particularly relevant when evaluation noise from an unobservable target reduces the power to discriminate (Clements et al. 2024; Degiannakis et al. 2022).

For quantile forecasts, Christoffersen (1998) establishes conditional coverage testing: a well-specified quantile forecast at level τ should produce violations at rate τ , and these violations should be serially independent. The quantile loss function (the check function used in estimation) also serves as a natural evaluation metric, penalizing over- and under-prediction asymmetrically according to τ (Lyócsa and Molnár 2021). Puke and Schweikert (2026) show that aligning the estimation and evaluation loss function produces significant forecast gains; since HAR-QR uses the check function for both estimation and evaluation, it is inherently coherent by construction.

3 Methodology

This section specifies the econometric framework for forecasting cocoa futures volatility at horizons from one day to twelve months. We first define the volatility measure, then specify the HAR-QR model and its ENSO-augmented extension, describe the benchmark models, and finally lay out the evaluation framework. Table 1 provides an at-a-glance summary of the five models before the detailed specifications follow.

3.1 Volatility Measure

We measure daily volatility using the Yang-Zhang estimator, which combines overnight and trading-session variance components:

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_O^2 + k\hat{\sigma}_C^2 + (1 - k)\hat{\sigma}_{RS}^2, \quad (1)$$

where $\hat{\sigma}_O^2$ captures overnight variance, $\hat{\sigma}_C^2$ captures open-to-close variance, and $\hat{\sigma}_{RS}^2$ is the Rogers-Satchell component that exploits the daily range. The weighting factor

Table 1: Overview of models compared in this study, ordered from simplest to most complex.

Feature	Benchmarks			Main models	
	Hist. Vol.	GARCH(1,1)	HAR-OLS	HAR-QR	HAR-X-QR
Model class	Non-param.	ARCH-type	Linear reg.	Quantile reg.	Quantile reg.
HAR lags ($d/w/m$)	No	No	Yes	Yes	Yes
ENSO regressor	No	No	No	No	Yes
Quantile output	Empirical	Simulation	None	Direct	Direct
Estimation	—	MLE	OLS	Check fn.	Check fn.
Multi-period design	Rolling window	Recursive	Direct	Direct	Direct
Thesis role	Naive BM	Param. BM	HAR benchmark	Main (RQ1)	Main (RQ2)

Note: Direct = horizon-specific model estimated by direct projection; Recursive = closed-form GARCH mean recursion, quantiles via Gaussian closed-form (Eq. 13); Check fn. = check-function (quantile) loss; BM = benchmark; Hist. Vol. = rolling 252-day window with empirical quantiles.

$k = 0.34/(1.34 + (n + 1)/(n - 1))$ minimizes estimator variance.

This choice is motivated by two considerations. First, Yang-Zhang achieves efficiency gains over close-to-close estimators and second, it can be computed over our full 40-year daily sample, unlike realized volatility which requires high-frequency data. We validate the Yang-Zhang proxy against five-minute realized volatility in Section 4, comparing rolling-window aggregates (22-day averages) of both measures via a Mincer-Zarnowitz regression. This comparison at the multi-day level aligns with the HAR model’s use of averaged volatility as both predictors and targets.

3.2 The HAR-QR Model

Combining the HAR framework with quantile regression yields the HAR-QR model. The HAR-QR model for horizon h and quantile τ is

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau)\sigma_t^{(d)} + \beta_w(\tau)\sigma_t^{(w)} + \beta_m(\tau)\sigma_t^{(m)}, \quad (2)$$

where $\mathcal{F}_t$ denotes the information set at time t . The volatility components capture variation at three frequencies:

$$\sigma_t^{(d)} = \sigma_t, \quad (3)$$

$$\sigma_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 \sigma_{t-i}, \quad (4)$$

$$\sigma_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} \sigma_{t-i}. \quad (5)$$

The key feature of quantile regression is that each quantile τ has its own coefficient vector, allowing the relationship between predictors and future volatility to differ across

the distribution. For the median ($\tau = 0.50$), the coefficients may resemble those from OLS estimation. For the upper quantile ($\tau = 0.95$), the coefficients capture how the predictors relate to volatility during high-volatility episodes. We report three quantiles: $\tau \in \{0.50, 0.75, 0.95\}$. The median $\tau = 0.50$ serves as an anchor to OLS-based point forecasts. The intermediate quantile $\tau = 0.75$ traces the transition from central to tail behaviour. The upper quantile $\tau = 0.95$ is of primary interest, as it captures the conditional upper bound of volatility, the regime where mean forecasts are most likely to understate actual outcomes. Lower quantiles ($\tau = 0.05, 0.25$) are omitted as the thesis focuses on upper-tail risk.

The dependent variable is the average daily volatility over the forecast horizon, following the HAR literature convention (Ghysels and Marcellino 2018; Clements et al. 2024):

$$\bar{\sigma}_{t:t+h} = \frac{1}{h} \sum_{i=1}^h \sigma_{t+i}. \quad (6)$$

This averaging ensures comparability across horizons and aligns with the HAR model's construction, where weekly and monthly components are themselves averages of daily volatility.

We adopt the *direct* forecasting approach: a separate model is estimated for each horizon h , using $\bar{\sigma}_{t:t+h}$ as the dependent variable. This contrasts with *iterated* forecasting, which estimates a one-step model and iterates forward. Ghysels and Marcellino (2018) find that direct forecasting is preferred for realized volatility models. For quantile regression, direct estimation is particularly natural, as it produces the conditional quantile at horizon h without the distributional assumptions required to iterate quantile forecasts. Our longest horizons (6–12 months) would require iterating hundreds of daily steps, amplifying any model misspecification.

We estimate HAR-QR at six horizons: $h \in \{1, 5, 22, 66, 132, 264\}$ trading days, corresponding to 1 day, 1 week, 1 month, 3 months, 6 months, and 12 months. The short horizons (1 day to 1 month) serve to validate the methodology against existing literature: HAR-QR has been tested at horizons up to 22 trading days (Lyócsa and Molnár 2021). The long horizons (6 and 12 months) constitute the core contribution, testing whether HAR-QR can produce useful forecasts at horizons where no prior study has ventured.

At long horizons, the autoregressive signal from lagged volatility fades. We therefore augment the HAR-QR model with the ENSO climate index:

$$Q_{\tau}(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau)\sigma_t^{(d)} + \beta_w(\tau)\sigma_t^{(w)} + \beta_m(\tau)\sigma_t^{(m)} + \gamma(\tau)\text{ENSO}_t, \quad (7)$$

where ENSO_t is the Niño 3.4 sea surface temperature anomaly index from NOAA, measured monthly. The coefficient $\gamma(\tau)$ captures the marginal effect of ENSO conditions

on future volatility at quantile τ .

We use *contemporary* ENSO, meaning ENSO_t predicts $\bar{\sigma}_{t:t+h}$, rather than explicitly lagged ENSO values. This specification is standard in the climate-commodity literature (Bouri et al. 2021; Bonato et al. 2023) and is fully causal: the Niño 3.4 anomaly for month t is publicly observable before any forecast is issued, so no forward-looking information enters the model. The sensitivity of this specification to alternative ENSO lags is examined in Section 5.5.

We obtain monthly Niño 3.4 sea surface temperatures from NOAA’s Extended Reconstructed SST (ERSSTv5) dataset and convert them to anomalies by subtracting the 1991–2020 climatological monthly mean, following standard practice (Bouri et al. 2021; Bonato et al. 2023). Positive anomalies indicate El Niño conditions; negative anomalies indicate La Niña. Since the index is monthly, each trading day inherits the Niño 3.4 anomaly of its calendar month.

The HAR-QR model is estimated by minimizing the check function loss:

$$\hat{\beta}(\tau) = \arg \min_{\beta} \sum_{t=1}^{T-h} \rho_{\tau}(\bar{\sigma}_{t:t+h} - X_t' \beta), \quad (8)$$

where $X_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)})'$ for the basic model, or $X_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)}, \text{ENSO}_t)'$ for the extended model. The check function is defined as:

$$\rho_{\tau}(u) = u \cdot (\tau - \mathbf{1}_{u < 0}) = \begin{cases} \tau \cdot u & \text{if } u \geq 0, \\ (\tau - 1) \cdot u & \text{if } u < 0. \end{cases} \quad (9)$$

This asymmetric loss function penalizes under-prediction more heavily when τ is large and over-prediction more heavily when τ is small, ensuring that the estimated quantile achieves the target coverage rate. The minimization is a linear programming problem solved efficiently using the simplex algorithm (Koenker and d’Orey 1987). Standard errors are computed using the bootstrap.

3.3 Benchmark Models

We compare HAR-QR against three benchmarks that span the major modeling paradigms discussed in the literature review.

The first benchmark is *rolling historical volatility* with a one-year window ($w = 252$):

$$\hat{\sigma}_{t+h}^{\text{hist}} = \sqrt{\frac{1}{w} \sum_{i=1}^w (r_{t-i+1} - \bar{r})^2}. \quad (10)$$

For quantile forecasts, we compute empirical quantiles of the historical volatility distribution: sort the w past volatilities and select the value at position $\lceil \tau \cdot w \rceil$. This model

captures only unconditional variation and serves as a naive baseline.

The second benchmark is the *GARCH(1,1)* model (Bollerslev 1986):

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (11)$$

For mean forecasts at horizon h , the *GARCH(1,1)* produces closed-form predictions:

$$\mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2), \quad (12)$$

where $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$ is the unconditional variance and $(\alpha+\beta)^{h-1}$ governs the speed of mean reversion. At long horizons, all h -step forecasts converge to $\bar{\sigma}^2$, irrespective of current conditions, a structural limitation of daily *GARCH* at 6–12 month horizons.

For quantile forecasts we adopt a closed-form Gaussian approach consistent with the normal-innovation assumption used in estimation. The mean forecast $\hat{\sigma}_{t,h}$ is obtained as the square root of the average h -step variance, $\hat{\sigma}_{t,h} = \sqrt{h^{-1} \sum_{k=1}^h \mathbb{E}_t[\sigma_{t+k}^2]}$. Forecast uncertainty is captured by the in-sample standard deviation s_t of daily volatility over the estimation window. The τ -quantile forecast is then

$$\hat{Q}_{\tau,t}^{\text{GARCH}} = \hat{\sigma}_{t,h} + \Phi^{-1}(\tau) s_t, \quad (13)$$

where $\Phi^{-1}(\tau)$ is the standard normal quantile function.

The third benchmark is *HAR-OLS*, the standard *HAR* model estimated by ordinary least squares. *OLS* yields only a conditional mean forecast $\hat{\mu}_{t,h}$, but a distributional forecast can be constructed from the in-sample residuals. Let $\hat{\varepsilon}_i = \sigma_i - \hat{\mu}_i$ denote the residuals from the training window. The τ -quantile forecast is then

$$\hat{Q}_{\tau,t}^{\text{OLS}} = \hat{\mu}_{t,h} + \hat{F}_{\varepsilon}^{-1}(\tau), \quad (14)$$

where $\hat{F}_{\varepsilon}^{-1}(\tau)$ is the empirical τ -quantile of the training-window residuals. This pseudo-quantile approach requires no distributional assumption and produces forecasts that are directly comparable to *HAR-QR*: if *HAR-QR* provides no improvement over *HAR-OLS* at quantile τ , then conditioning on the *HAR* structure suffices and quantile regression adds no value. Comparing the two at $\tau = 0.50$ tests symmetry of the conditional distribution; comparing at $\tau = 0.95$ tests whether the upper tail is better captured by explicit quantile estimation.

3.4 Evaluation Framework

Evaluating volatility forecasts is complicated by the fact that volatility is unobserved, and every evaluation compares forecasts against an imperfect proxy (Patton 2011). We evaluate forecasts against Yang-Zhang volatility computed from realized prices, using

loss functions that Patton (2011) has shown to be robust to noise in the volatility proxy.

For mean and median forecasts, we use QLIKE as the primary metric:

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^T \left(\log \hat{\sigma}_t^2 + \frac{\sigma_t^2}{\hat{\sigma}_t^2} \right). \quad (15)$$

Patton (2011) derives necessary and sufficient conditions for the loss function to yield forecast rankings that are robust to proxy noise, showing that among nine commonly used loss functions, only MSE and QLIKE satisfy these conditions. We prefer QLIKE because it penalizes under-prediction of volatility proportionally more than MSE, which is appropriate when under-prediction is costlier than over-prediction, a property particularly relevant for upper-quantile forecasts. As Puke and Schweikert (2026) demonstrate, aligning estimation and evaluation loss functions, what they call *coherent forecasting*, yields significant accuracy gains. Our HAR-QR models are coherent by construction: the check function (Equation 9) serves as both the estimation objective and the evaluation metric.

As a robustness check, we additionally report mean squared error (MSE) and mean absolute error (MAE) for all point-forecast models:

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T (\sigma_t - \hat{\sigma}_t)^2, \quad (16)$$

$$\text{MAE} = \frac{1}{T} \sum_{t=1}^T |\sigma_t - \hat{\sigma}_t|. \quad (17)$$

MSE penalizes large errors quadratically and is a member of the proxy-robust class alongside QLIKE (Patton 2011); including it confirms that QLIKE rankings are not sensitive to the choice of loss function within that class. MAE penalizes all deviations linearly and is not proxy-robust, but provides an interpretable measure of average forecast error in the original volatility units.

For quantile forecasts, we use quantile loss:

$$\text{QL}_\tau = \frac{1}{T} \sum_{t=1}^T \rho_\tau(\sigma_t - \hat{Q}_{\tau,t}). \quad (18)$$

As a complementary diagnostic, we assess the calibration of quantile forecasts through their *empirical violation rate*. For a given quantile level τ , the violation rate measures the fraction of out-of-sample observations that exceed the quantile forecast:

$$\widehat{VR}_\tau = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\sigma_t > \hat{Q}_{\tau,t}\}, \quad (19)$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. A well-calibrated τ -quantile forecast should satisfy $\widehat{VR}_\tau \approx 1 - \tau$. For example, the $\tau = 0.95$ quantile should be exceeded approximately 5% of the time; systematic departures indicate miscalibration. Violation rates above $1 - \tau$ signal that the model underestimates the quantile (too narrow), while rates below $1 - \tau$ indicate overestimation (too conservative). This diagnostic complements quantile loss, which measures average forecast accuracy but does not directly reveal whether the nominal coverage level is achieved.

To determine whether differences in forecast accuracy are statistically significant, we use the Model Confidence Set (MCS) of Hansen et al. (2011). The MCS identifies the set of models $\mathcal{M}^*$ that contains the best model with a given confidence level, avoiding the need to specify a single benchmark. Starting from the full collection of competing models, the procedure iteratively tests for equal predictive ability and eliminates the worst-performing model until non-rejection occurs. We implement MCS with 10,000 stationary bootstrap replications using the max- t statistic; models in $\mathcal{M}^*$ at the 90% confidence level are considered statistically equivalent.

The forecast evaluation is conducted out-of-sample using a rolling estimation window of 2,520 trading days (approximately 10 years). At each forecast origin t , we estimate parameters using observations from $t - 2519$ to t , generate forecasts, and roll forward by one day. The window rolls *daily* to maximize the number of forecast–realization pairs for statistical evaluation. With approximately 30 years of out-of-sample data, this yields thousands of pairs for statistical testing.

Several assumptions underpin the methodology. First, we assume *local stationarity*: volatility dynamics are stable within each estimation window, though they may shift across windows. The rolling-window design accommodates gradual parameter changes. Second, we assume that the Yang-Zhang estimator provides a reasonable proxy for latent volatility, which we validate empirically in Section 4. Third, we maintain *linear conditional quantiles* for parsimony, acknowledging that non-linear extensions could potentially improve forecasts. Fourth, we assume ENSO is exogenous to the cocoa market, which is physically reasonable.

Based on the literature, we expect that at short horizons (1 day to 1 month), HAR-QR will match or outperform benchmarks. At long horizons (6–12 months), standard HAR-QR will lose predictive power; we test whether HAR-X-QR augmented with ENSO recovers predictive value, examining whether climate information provides incremental value precisely where the autoregressive signal fades.

Robustness diagnostics for the ENSO signal. Should HAR-X-QR outperform HAR-QR at long horizons, a single full-sample comparison cannot by itself establish whether the gain reflects a structural climate channel or a sample-specific coincidence. We therefore subject any ENSO improvement to three diagnostics, reported in Sec-

tion 5.5. First, a *temporal-stability* check re-evaluates the gain on a pre-crisis subsample ending 8 February 2024, estimated via a sup-Wald structural break test (Andrews 1993), which excludes the extreme-price phase of the 2022–2024 supply crisis: a structural relationship should survive the exclusion of any single episode. Second, a *lag-structure* check re-estimates HAR-X-QR with the Niño 3.4 index lagged 0, 1, 3, and 6 months and inspects the cross-correlogram between ENSO and cocoa volatility. Under the agronomic-transmission hypothesis the predictive content should strengthen at lags of 6–12 months, matching the window over which ENSO-driven rainfall anomalies are hypothesised to affect harvests; this is the lag horizon at which ENSO is found to improve commodity volatility forecasts in the oil and agricultural literature (Bouri et al. 2021; Bonato et al. 2023). The cross-correlogram follows standard cross-correlation diagnostics for lead–lag relationships between time series (Tsay 2010). Third, a *cross-commodity* check applies the identical HAR-X-QR specification to Arabica coffee and raw sugar—two tropical soft commodities that share weather exposure and downstream markets with cocoa (Ge et al. 2025)—since a genuine climate channel should produce comparable gains across commodities grown in the same equatorial belt (Ubilava 2018; Bonato et al. 2023).

4 Data

This section describes the data used to estimate and evaluate the volatility forecasting models developed in Section 3. Two primary datasets are required: daily futures prices for ICE London cocoa contracts and the Niño 3.4 climate index measuring El Niño–Southern Oscillation (ENSO) activity. After describing the cocoa futures data, including contract specifications, the construction of a continuous price series, and the variables derived from it, we turn to the ENSO index used as an exogenous predictor at long horizons. The section concludes with descriptive statistics documenting the stylized facts of cocoa returns and volatility.

4.1 Cocoa Futures Data

The primary data consist of daily Open-High-Low-Close (OHLC) prices and trading volume for ICE Futures Europe cocoa futures contracts. ICE London cocoa futures (contract code LCC) are standardized contracts for the delivery of 10 metric tonnes of cocoa beans. Each contract specifies delivery in one of five months: March, May, July, September, or December. Prices are quoted in British pounds sterling (GBP) per metric tonne, and trading occurs on the IntercontinentalExchange (ICE) electronic platform during European market hours. The data were obtained from Refinitiv Workspace via the corporate license held by Ritter Sport GmbH & Co. KG, ensuring institutional-

grade data quality sourced directly from exchange feeds.

Continuous series construction. Futures contracts have finite maturity, requiring the construction of a continuous price series for volatility analysis. We use the generic front-month continuous contract, denoted LCCc1 in the Refinitiv system. This series rolls to the next available contract approximately one month before expiration, maintaining exposure to the most liquid contract while avoiding delivery. The front-month contract typically exhibits the highest trading volume and the tightest bid-ask spreads, making it the standard choice for volatility research on commodity futures (Degianakis et al. 2022; Alfeus and Nikitopoulos 2022). Rolling between contracts creates artificial price jumps when the new contract trades at a different level than the expiring one (the roll yield). For volatility estimation using OHLC data, these jumps are largely captured within the daily range and do not require explicit adjustment. The Yang-Zhang estimator (Section 3.1) is specifically designed to be robust to opening jumps, which include both overnight news and contract roll effects.

Sample period. The daily dataset spans 13,090 trading days from April 1, 1974, to December 15, 2025. This extended sample offers several advantages. It encompasses multiple El Niño and La Niña cycles, providing sufficient variation in the ENSO predictor for long-horizon models. It includes diverse market regimes: the 1977–1979 cocoa price spike driven by supply disruptions in West Africa, the long period of relative stability from 1990–2010, the financialization era beginning around 2008, and the recent 2022–2024 price surge attributed to the Cocoa Swollen Shoot Virus Disease (CSSVD) outbreak. The sample also includes approximately one year of high-frequency data (February 3, 2025, through January 12, 2026, comprising 20,791 five-minute price observations), which enables validation of the Yang-Zhang volatility estimator against realized volatility computed from intraday returns.

Table 14 in Appendix A summarizes the coverage and key characteristics of the cocoa futures data.

Variables and transformations. For each trading day t , the dataset provides five price observations: the opening price O_t , the highest price during the trading session H_t , the lowest price L_t , the closing price C_t , and total trading volume V_t . Trading volume is collected but not used in the volatility models. Log returns $r_t = \ln(C_t/C_{t-1})$ are computed for descriptive statistics and as inputs to the GARCH benchmark.

From the daily OHLC prices, we construct volatility estimates using the Yang-Zhang estimator detailed in Equation (1). The estimator combines overnight variance (close-to-open), intraday range-based variance (Rogers-Satchell), and close-to-close variance into a minimum-variance unbiased estimator that is drift-independent

and robust to opening gaps (Yang and Zhang 2000). The Yang-Zhang estimator serves as the dependent variable in all volatility forecasting models and as the realized measure against which forecasts are evaluated.

For the one-year period with available intraday data, we compute realized volatility as the sum of squared five-minute log returns:

$$RV_t = \sum_{i=1}^M r_{t,i}^2, \quad (20)$$

where $r_{t,i}$ is the log return for the i -th five-minute interval on day t , and M is the number of five-minute intervals in the trading day (approximately 78 intervals for a 6.5-hour trading session). This realized volatility measure provides a benchmark for validating the Yang-Zhang estimator over the recent subsample.

Data quality. The daily data contain no missing values for trading days; non-trading days are excluded. Overnight returns spanning weekends or holidays are not rescaled by calendar days elapsed, as the larger return reflects genuine accumulated risk rather than a measurement artefact.

4.2 ENSO Climate Data

As discussed in Section 2.5, ENSO is hypothesised to provide a physically motivated predictor for cocoa volatility at long horizons. We measure ENSO activity using the Niño 3.4 index, a measure of sea surface temperature anomalies in the central equatorial Pacific (5°N–5°S, 170°W–120°W), computed as a three-month running mean relative to a climatological base period. Monthly values are obtained from the NOAA Climate Prediction Center, calculated using the ERSSTv5 dataset and publicly available at <https://psl.noaa.gov/data/correlation/nina34.data>.

Our sample covers January 1960 through July 2025, providing 786 monthly observations. NOAA classifies an El Niño (La Niña) episode when the three-month running mean exceeds +0.5°C (falls below −0.5°C) for at least five consecutive months. In the regression models (Equation 7), $ENSO_t$ enters as the continuous Niño 3.4 value rather than a categorical regime indicator, preserving the full variation in anomaly magnitude.

The sample contains 15 El Niño and 13 La Niña episodes. Since the index is monthly while cocoa futures trade daily, each trading day inherits the Niño 3.4 value of its calendar month, as described in Section 3.2. The data contain no missing values through June 2025; placeholder entries (−99.99) for August 2025 onward do not affect the analysis because the out-of-sample period ends before these values are needed.

Table 2 summarizes the Niño 3.4 data.

Table 2: Summary of Niño 3.4 Climate Data

Characteristic	Value
Measures	Sea surface temperature anomaly
Region	Central equatorial Pacific (5°N–5°S, 170°W–120°W)
Data source	NOAA Climate Prediction Center
Dataset	ERSSTv5
URL	https://psl.noaa.gov/data/correlation/nina34.data
Sample period	January 1960 – July 2025
Observations	786 months
Frequency	Monthly
El Niño threshold	$> +0.5^{\circ}\text{C}$ (5 consecutive months)
La Niña threshold	$< -0.5^{\circ}\text{C}$ (5 consecutive months)

4.3 Descriptive Statistics and Stylized Facts

This subsection presents the empirical characteristics of cocoa returns and volatility, establishing the stylized facts that motivate the modeling choices in Section 3.

Price and return characteristics. Figure 1 displays the daily closing price of the front-month cocoa futures contract over the full sample period. The series exhibits several distinct episodes. The late 1970s witnessed a pronounced price spike, with cocoa reaching approximately 3,400 GBP per tonne in 1977–1979, driven by adverse weather and supply disruptions in West Africa. Following this episode, prices stabilized at lower levels through the 1990s and 2000s, with moderate fluctuations around a mean of approximately 1,300 GBP per tonne. The 2010s showed gradual price increases, followed by the recent 2022–2024 surge reaching over 10,000 GBP per tonne, unprecedented in the modern trading history of cocoa, driven by the CSSVD outbreak in key producing regions.

Table 3 presents descriptive statistics for daily log-returns. The mean daily return is slightly positive (0.012%), corresponding to an approximate annualized mean return of 3% over the full sample. However, returns exhibit substantial volatility: the standard deviation of 1.93% implies an annualized volatility of approximately 30%, consistent with cocoa’s classification as a high-volatility commodity. The distribution of returns deviates sharply from normality. Negative skewness (-0.997) indicates a left tail longer than the right, reflecting the tendency for large negative price movements to occur more frequently than large positive ones. Excess kurtosis of 12.96 documents fat tails: extreme returns occur far more often than a Gaussian distribution would predict. The Jarque-Bera test decisively rejects normality ($p < 0.001$), confirming that standard volatility models assuming Gaussian innovations may be misspecified for cocoa returns.

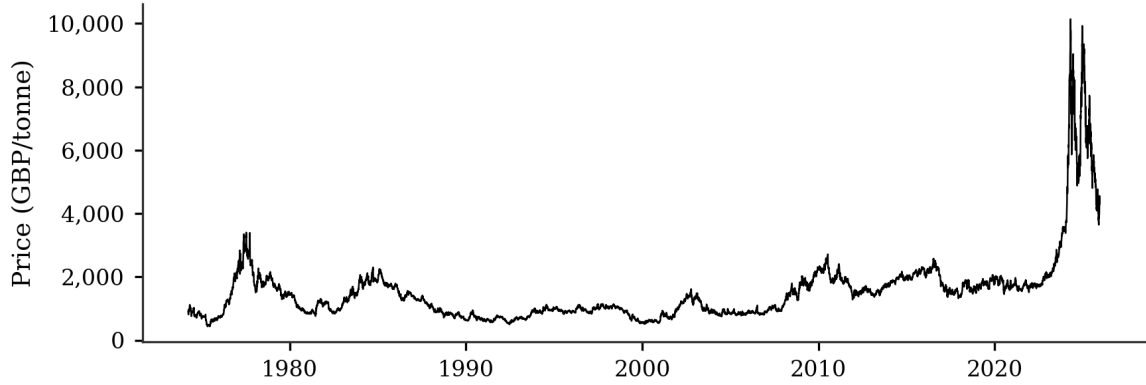


Figure 1: ICE London Cocoa Futures Daily Closing Price (1974–2025)

Notes: Front-month continuous contract (LCCc1) daily closing prices in GBP per metric tonne. Data source: Refinitiv Workspace. Sample period: April 1, 1974 – December 15, 2025 (13,090 trading days).

Table 3: Descriptive Statistics for Daily Log>Returns

Statistic	Value
Observations	13,087
Mean (%)	0.012
Median (%)	0.000
Std. Deviation (%)	1.929
Minimum (%)	−26.212
Maximum (%)	12.988
25th Percentile (%)	−0.917
75th Percentile (%)	0.945
Skewness	−0.997
Excess Kurtosis	12.963
Jarque-Bera	141,013
<i>p</i> -value	< 0.001

Notes: Daily log-returns computed as $r_t = \ln(C_t/C_{t-1})$ for ICE London cocoa front-month continuous contract (LCCc1), April 1974–December 2025. Returns expressed in percentage points. Excess kurtosis reported (normal distribution: 0). Jarque-Bera test strongly rejects normality ($p < 0.001$), consistent with fat tails and negative skewness typical of commodity returns.

Volatility time series. Figure 2 displays the Yang-Zhang volatility estimator (annualized, in percent) computed from daily OHLC data. The figure reveals pronounced volatility clustering: periods of high volatility persist for extended intervals before reverting to lower levels. The late-1970s episode and the 2022–2024 crisis stand out as the two dominant volatility regimes in the sample. Between these episodes, volatility fluctuates around a lower baseline with occasional spikes corresponding to supply shocks, weather events, and financial market turbulence (e.g., the 2008 financial crisis).

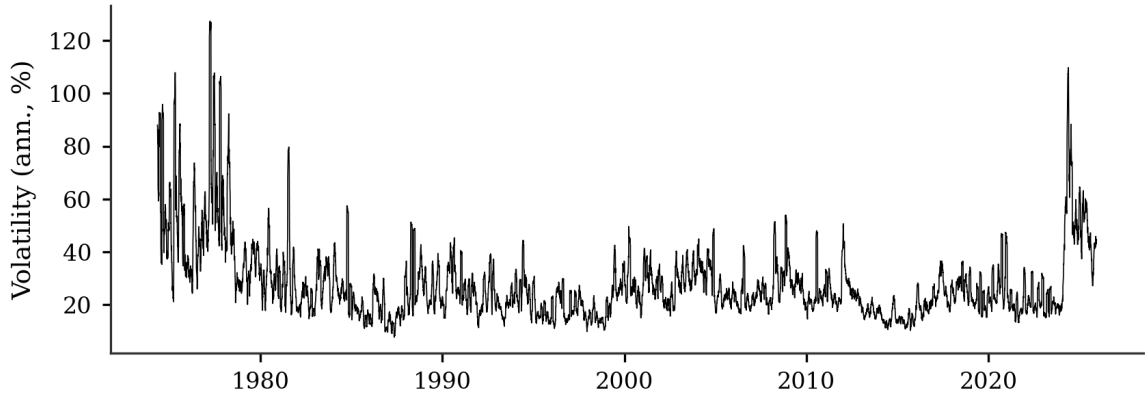


Figure 2: Yang-Zhang Annualized Volatility Time Series (1974–2025)

Notes: Yang-Zhang volatility estimator computed from daily OHLC prices and annualized (percent). The estimator combines overnight, intraday, and close-to-close variance components following Yang and Zhang (2000). Pronounced volatility clustering and regime-switching are evident, motivating the use of HAR models that capture persistence across multiple horizons.

The persistence of volatility justifies the HAR model’s multi-horizon structure: current volatility depends not only on yesterday’s volatility (daily component) but also on average volatility over the past week and month (weekly and monthly components). The clustering pattern also motivates quantile regression: during high-volatility regimes, the tails of the volatility distribution shift outward, and point forecasts of the mean may underestimate the risk of extreme outcomes.

Stationarity. The regression models in Section 3 require stationary inputs. Table 15 reports Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) unit root tests for the three key series (reported in Table 15, Appendix A). Both tests include a constant; ADF lag length is selected by AIC.

All three series decisively reject the unit root null at the 1% level. Log returns are stationary by construction, with ADF and PP statistics both below -112 . The Yang-Zhang volatility series exhibits high persistence (the ADF test requires 39 lags to whiten the residuals) but remains stationary ($\text{ADF} = -8.42$, $p < 0.001$). The large discrepancy between the ADF and PP statistics (-8.42 vs. -90.99) reflects the difference in how each test handles serial correlation: ADF augments the regression with

lagged differences, while PP applies a non-parametric correction to the test statistic. Both agree on rejection. The Niño 3.4 anomaly is stationary by design, as it measures deviations from a climatological mean, confirmed by both tests (ADF = -6.85 , PP = -6.46 , $p < 0.001$). These results justify the use of standard OLS and quantile regression methods without differencing or cointegration corrections.

The data described in this section provide the foundation for the empirical analysis. The extended daily sample (1974–2025) ensures sufficient variation in both volatility regimes and ENSO phases, while the availability of intraday data for the recent period enables validation of the Yang-Zhang volatility proxy. That validation, reported in Table 17 of Appendix A, confirms that the Yang-Zhang estimator tracks intraday realized volatility closely, with the correlation rising to 0.92 at the monthly frequency relevant to the forecasting exercise. The next section presents the empirical results from the HAR-based quantile regression models estimated using these data.

5 Empirical Results

This section presents the out-of-sample forecasting results, organized around the two research questions. Section 5.1 compares mean forecast accuracy across the three baseline models. Section 5.2 evaluates distributional forecasts at the upper tail, testing whether HAR-QR outperforms the benchmarks across horizons (RQ1). Section 5.3 assesses whether the Niño 3.4 climate index adds forecast value at long horizons (RQ2). Section 5.4 tests the statistical significance of the observed differences using Model Confidence Sets. Section 5.6 synthesizes the findings. All results use a rolling estimation window of 2,520 trading days with daily re-estimation, yielding approximately 10,500 out-of-sample forecast–realization pairs at each horizon.

5.1 Mean Forecast Comparison

Table 4 reports the QLIKE loss for the three mean forecast models across all six horizons. The ranking is unambiguous: HAR-OLS achieves the lowest (best) QLIKE at every horizon, followed by GARCH(1,1), with historical volatility uniformly last.

Table 4: Mean forecast QLIKE loss by model and horizon

Model	1d	1w	1m	3m	6m	12m
Historical Vol	−7.002	−7.256	−7.327	−7.334*	−7.324	−7.310
GARCH(1,1)	−7.242	−7.388	−7.406*	−7.381	−7.356	−7.345
HAR-OLS	−7.302	−7.482	−7.521*	−7.515	−7.492	−7.485

Note: QLIKE loss (Patton 2011); more negative values indicate better forecast performance. Bold: best model at each horizon. *Best horizon for each model. Evaluation period: approximately 10,500 out-of-sample trading days (1984–2025).

The gap between HAR-OLS and GARCH is largest at the one-month horizon (0.115 QLIKE units) and narrowest at one day (0.060 units). This pattern reflects the structural differences between the two models. GARCH(1,1) captures short-run volatility dynamics effectively: its conditional variance process adapts quickly to recent shocks. But its exponential memory decay means that multi-step forecasts converge rapidly toward the unconditional variance, erasing the model’s informational advantage at longer horizons. HAR-OLS, by contrast, includes a monthly volatility component ($\sigma_t^{(m)}$) that captures persistence at lower frequencies. This component maintains a correlation of 0.66 with the twelve-month target, compared to 0.37 for the daily component, explaining why HAR-OLS retains a substantial edge even at twelve months.

Historical volatility performs worst because it is the most constrained: its trailing average cannot distinguish between a volatility spike driven by a transient shock and a persistent regime change. Nevertheless, all three models share a common horizon pattern: loss improves from the one-day to the one-month horizon, reflecting the smoothing effect of averaging in the target $\bar{\sigma}_{t:t+h}$, before degrading at six and twelve months as the autoregressive signal weakens. The fact that even HAR-OLS degrades beyond one month confirms the fundamental difficulty of long-horizon volatility forecasting and motivates the search for additional predictive signals in Section 5.3.

The QLIKE ranking is confirmed by MSE and MAE. Figure 3 plots all three loss functions across horizons. HAR-OLS achieves the lowest MSE at every horizon; MAE is minimized by HAR-X-QR (median) at short horizons and by HAR-OLS or HAR-QR (median) beyond three months. Crucially, GARCH(1,1) is the worst-performing model under all three metrics at long horizons, with its error rising steeply relative to the HAR family as the horizon extends, a direct consequence of mean-reversion in its multi-step forecasts. The HAR family dominates regardless of how point forecast errors are penalized.

5.2 Quantile Forecast Comparison (RQ1)

Mean forecasts summarize only the center of the conditional distribution. For risk assessment, the upper tail matters: a model that accurately forecasts average volatility may still understate the probability of extreme outcomes. We now compare five models on their ability to forecast the 95th percentile of future volatility. Historical volatility uses the empirical quantile of the trailing 252-day distribution. GARCH(1,1) quantiles follow the closed-form Gaussian approach (Equation 13). HAR-OLS produces pseudo-quantile forecasts from its in-sample residual distribution (Equation 14). HAR-QR estimates the quantile directly through quantile regression on the HAR structure, and HAR-X-QR augments this with the Niño 3.4 index.

Table 5 reports the quantile loss at $\tau = 0.95$ for all five models. Figure 4 presents

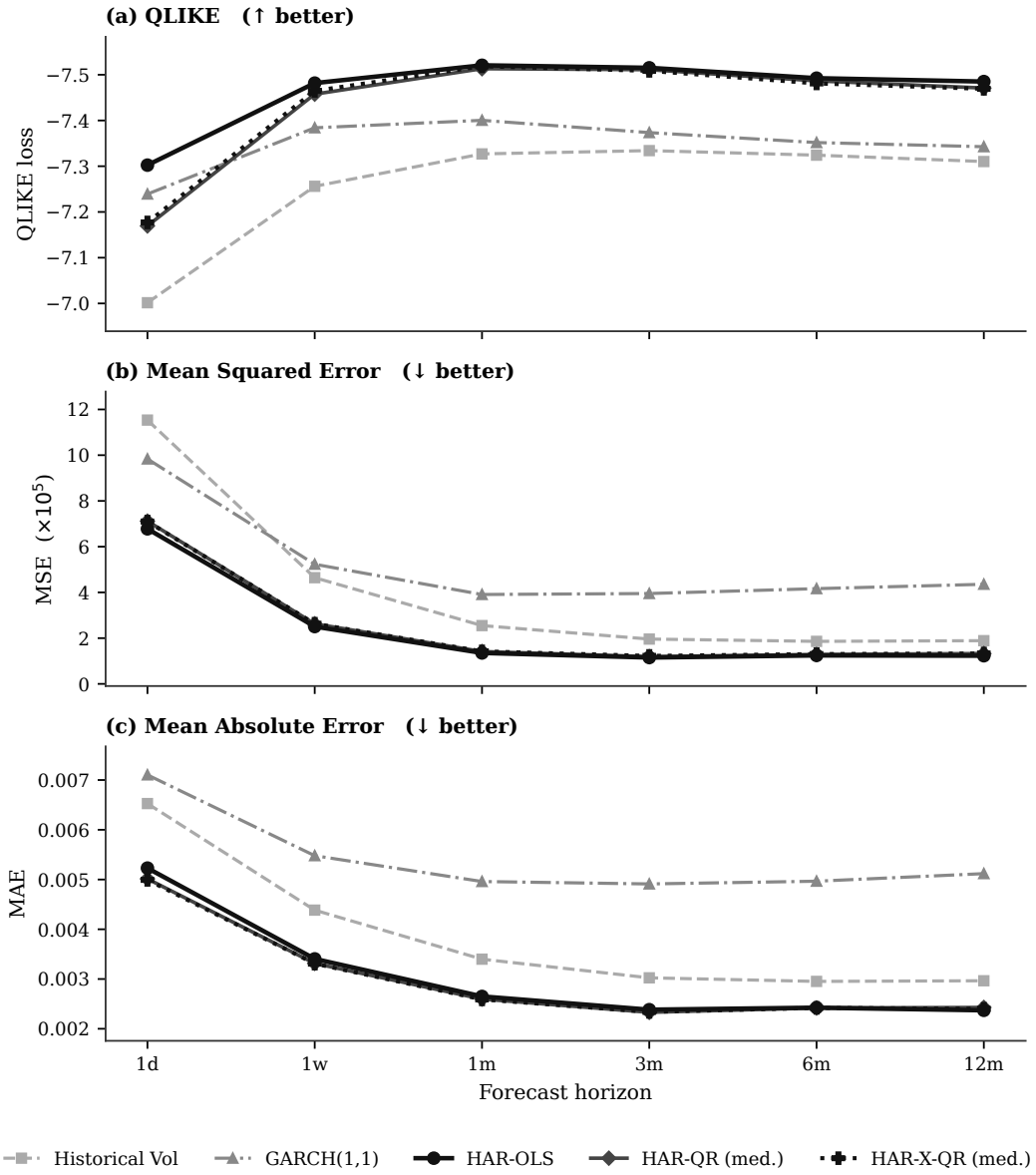


Figure 3: Robustness check: point forecast accuracy under three loss functions
Notes: Each panel plots mean (or median) forecast loss across horizons. QLIKE axis is inverted (more negative = better). MSE scaled by 10^5 for readability. All models evaluated on the same out-of-sample period. The consistent separation between GARCH(1,1) and the HAR family at long horizons holds across all three metrics.

the same comparison graphically.

Table 5: Upper-tail quantile loss ($\tau = 0.95$) by model and horizon

Model	1d	1w	1m	3m	6m	12m
Historical Vol	0.001502	0.001023	0.000871*	0.000889	0.000914	0.000930
GARCH(1,1)	0.001393	0.001067	0.001011*	0.001053	0.001066	0.001044
HAR-OLS	0.001272	0.000748	0.000507	0.000472	0.000479	0.000451*
HAR-QR	0.001193	0.000692	0.000458	0.000433*	0.000454	0.000446
HAR-X-QR	0.001195	0.000695	0.000464	0.000418	0.000387	0.000333*

Note: Quantile loss as defined in Equation (18). Lower values indicate better forecasts. Bold: best model at each horizon. *Best horizon for each model. Historical Vol quantiles are empirical percentiles of the trailing 252-day volatility distribution. GARCH quantiles use the closed-form Gaussian approach (Equation 13). HAR-OLS quantiles are pseudo-quantile forecasts from in-sample residuals (Equation 14).

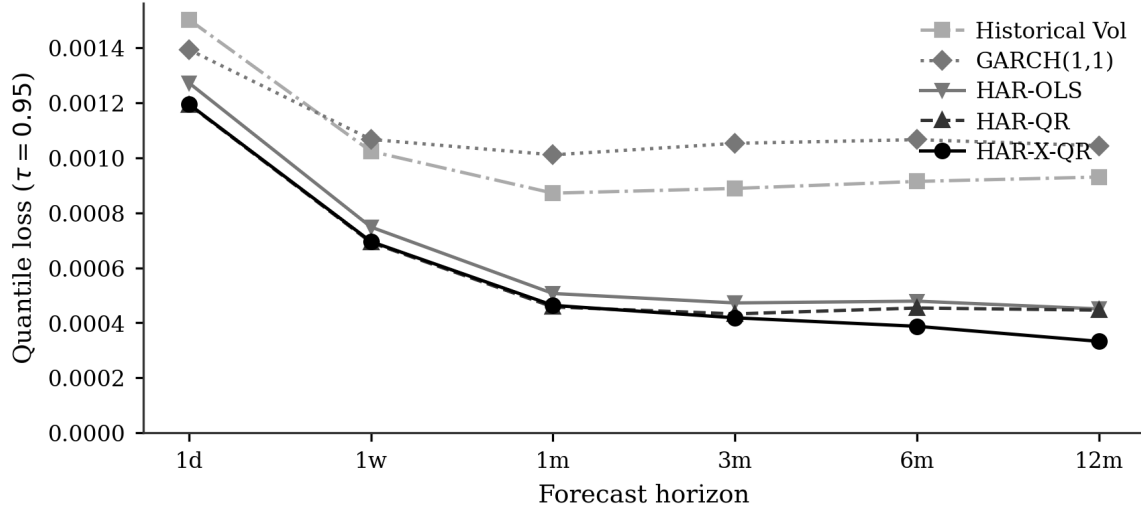


Figure 4: Upper-tail quantile loss ($\tau = 0.95$) across forecast horizons

Notes: Lower values indicate better upper-tail forecasts. The HAR family (HAR-OLS, HAR-QR, HAR-X-QR) dominates both non-parametric and GARCH benchmarks at all horizons. HAR-X-QR (ENSO-augmented) separates from HAR-QR at horizons beyond three months. The y-axis starts at zero for honest visual comparison.

Four patterns emerge from this comparison.

First, the HAR family dominates all benchmarks at every horizon. HAR-QR, HAR-OLS, and HAR-X-QR all outperform both GARCH(1,1) and historical volatility at every horizon. The gap is not marginal: at twelve months, the best benchmark (historical volatility, 0.000930) is 108% above HAR-QR (0.000446) and 179% above HAR-X-QR (0.000333). The HAR structure—which conditions on heterogeneous volatility lags at daily, weekly, and monthly frequencies—captures distributional persistence that neither the rolling unconditional distribution (historical volatility) nor the parametric GARCH model can exploit at procurement-relevant horizons.

Second, direct quantile estimation outperforms the pseudo-quantile approach. HAR-QR beats the HAR-OLS pseudo-quantile at every horizon, with improvements of 6% at one day (0.001193 vs 0.001272), 10% at one month (0.000458 vs 0.000507), and 1% at twelve months (0.000446 vs 0.000451). The gap is widest at short-to-medium horizons, where volatility dynamics are more state-dependent and the conditional quantile diverges from the unconditional residual distribution. At twelve months, the near-convergence of the two HAR variants indicates that the HAR regressors themselves carry most of the distributional information at very long horizons; the marginal contribution of re-weighting via quantile regression is modest once the ENSO regressor is absent. This confirms that the HAR-OLS pseudo-quantile is a useful benchmark: the advantage of explicit quantile regression is real but bounded.

Third, GARCH's Gaussian closed-form systematically underestimates extreme volatility beyond the short horizon. At one day, GARCH (0.001393) outperforms historical volatility (0.001502), reflecting the GARCH model's ability to condition the forecast distribution on the current volatility state. From one week onward, however, GARCH's Gaussian quantile assumption deteriorates. The closed-form formula uses the in-sample standard deviation of Yang-Zhang volatility as forecast uncertainty (Equation 13), which is a static measure that cannot adapt to the non-Gaussian, right-skewed shape of the volatility distribution at longer horizons. GARCH consequently places the $\tau = 0.95$ boundary too low, leading to excess violations and the highest quantile loss of all five models from one week onward. By twelve months, GARCH's quantile loss (0.001044) is 134% above HAR-QR's (0.000446).

Fourth, HAR-X-QR separates from HAR-QR only at long horizons. At one day through one month, the two HAR-QR models are virtually indistinguishable (differences $< 3\%$). The divergence begins at three months (+3.26% improvement) and accelerates to +14.66% at six months and +25.44% at twelve months. This horizon dependence is analyzed in detail in Section 5.3.

These findings answer RQ1: the HAR-QR model forecasts the upper tail of the conditional volatility distribution more accurately than all three benchmarks (historical volatility, GARCH(1,1), and HAR-OLS pseudoquantile) at all horizons from one day to twelve months.

Figures 5 and 6 illustrate this comparison visually. Each figure overlays the $\tau = 0.95$ forecast boundary from all four models on realized volatility for the 2015–2025 window, which includes the 2022–2024 cocoa supply crisis.

At the one-month horizon (Figure 5), the HAR-QR and HAR-X-QR boundaries track the realized upper tail closely, rising in tandem with the crisis spike. GARCH's boundary reacts with a visible lag, and historical volatility's boundary is the slowest to adjust. At the twelve-month horizon (Figure 6), the differences are more pronounced: the HAR-X-QR boundary rises ahead of the crisis period, reflecting the ENSO regres-

sor's contribution, while the GARCH boundary remains comparatively flat, a direct consequence of its mean-reverting multi-step forecasts converging toward the unconditional variance.

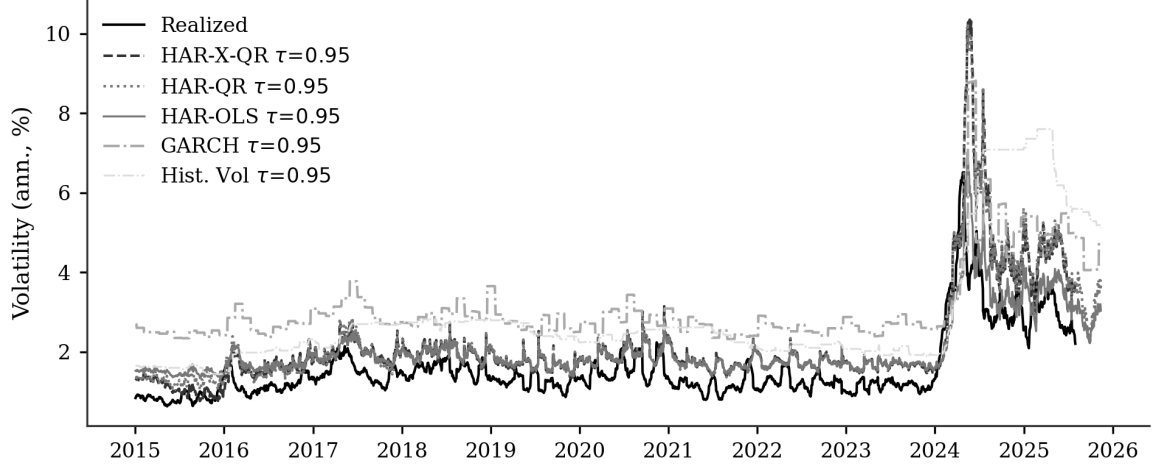


Figure 5: Upper-tail ($\tau = 0.95$) forecast comparison at the one-month horizon

Notes: Realized Yang-Zhang volatility (solid black) and $\tau = 0.95$ quantile forecasts from the five models. Zoomed to 2015–2025 to show the 2022–2024 supply crisis period in detail.

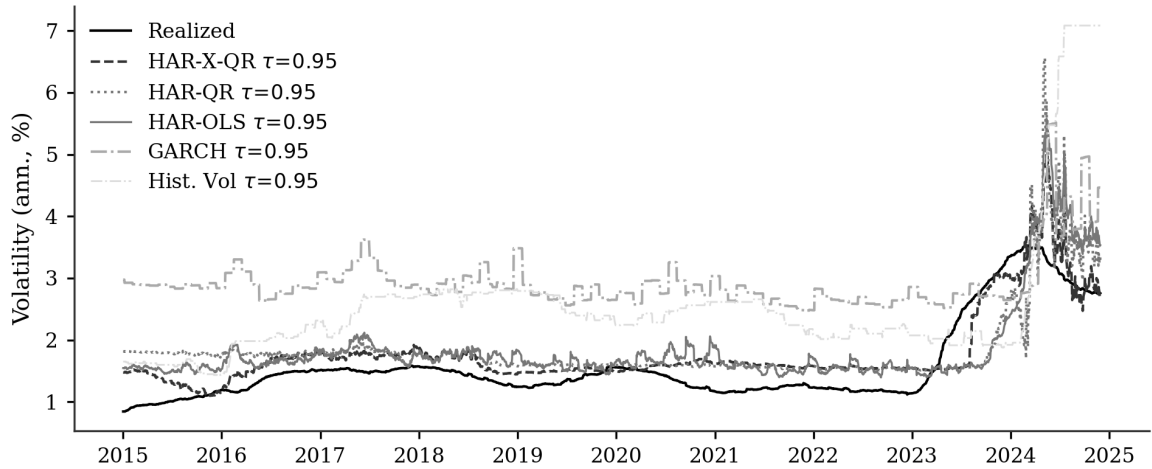


Figure 6: Upper-tail ($\tau = 0.95$) forecast comparison at the twelve-month horizon

Notes: Same construction as Figure 5. At the twelve-month horizon, HAR-X-QR's boundary rises ahead of the crisis period, while GARCH's boundary remains comparatively flat due to mean-reversion in its multi-step variance forecast and the Gaussian tail assumption.

Table 6 reports violation rates for the HAR-QR upper quantile ($\tau = 0.95$) by horizon. Under correct conditional coverage, the violation rate should equal $1 - \tau = 5\%$. At the one-day and one-week horizons, the model achieves near-target coverage (4.84% and 5.15%). Coverage remains acceptable through the three-month horizon (5.90%), but exceeds the target at six months (7.04%) and twelve months (8.34%). The coverage

degradation at long horizons indicates that the autoregressive signal alone becomes insufficient to track extreme upper-tail volatility: the HAR structure, while dominant among the models tested, still underestimates how far the distribution can shift in periods of sustained supply stress. This motivates the ENSO augmentation examined next.

Table 6: HAR-QR upper-quantile ($\tau = 0.95$) coverage by horizon

Horizon	N	Violations	Violation rate (%)
1d	10,545	510	4.84
1w	10,541	543	5.15
1m	10,524	600	5.70
3m	10,480	618	5.90
6m	10,414	733	7.04
12m	10,282	857	8.34

Note: A violation occurs when realized volatility exceeds the $\tau = 0.95$ quantile forecast. Under correct specification, the violation rate should equal 5%. N denotes the number of forecast–realization pairs at each horizon.

5.3 ENSO Augmentation: HAR-X-QR (RQ2)

Table 7 reports the percentage change in quantile loss when moving from HAR-QR to HAR-X-QR across all three quantile levels and six horizons. Positive values indicate improvement (lower loss under HAR-X-QR).

Table 7: Quantile loss improvement (%) from ENSO augmentation (HAR-X-QR vs. HAR-QR)

τ	1d	1w	1m	3m	6m	12m
0.50	−0.0076	−0.2093	−0.2698	−0.3994	−0.0643	+1.1225
0.75	−0.0357	−0.1810	−0.6684	−1.3689	−0.5717	−0.8660
0.95	−0.1628	−0.5111	−1.4628	+3.2564	+14.6602	+25.4375

Note: Positive values indicate that HAR-X-QR achieves lower quantile loss than HAR-QR. Computed as $(1 - \text{QL}_{\text{HAR-X}}/\text{QL}_{\text{HAR}}) \times 100$.

The results reveal a striking horizon- and quantile-dependent pattern. At short horizons (one day to one month), ENSO augmentation provides no benefit and occasionally worsens forecasts marginally. At such horizons, the Niño 3.4 coefficient captures noise rather than signal, and the additional parameter slightly degrades out-of-sample performance.

The first meaningful improvement appears at three months for the upper tail: $\tau = 0.95$ quantile loss decreases by 3.26%. The effect strengthens substantially at six months (+14.66%) and reaches its maximum at twelve months (+25.44%). This concentration in the upper tail is consistent with the hypothesis that ENSO anomalies

are associated with cocoa supply disruptions, which manifest as rightward shifts in the volatility distribution rather than a uniform upward shift. Whether this recalibration reflects a stable predictive relationship or a sample-specific regime effect is assessed in Section 5.5. The GARCH and historical volatility benchmarks cannot exploit this climate index, explaining why their twelve-month upper-tail performance in Table 5 falls far behind HAR-X-QR: historical volatility’s loss (0.000930) is 179% higher than HAR-X-QR’s (0.000333), and GARCH’s (0.001044) is 213% higher.

The median ($\tau = 0.50$) shows only modest improvement (+1.12%), confirming that the benefit of ENSO augmentation is predominantly distributional rather than centered on the mean.

Table 8 examines how ENSO augmentation affects upper-tail ($\tau = 0.95$) coverage calibration. HAR-X-QR achieves substantially lower quantile loss but its violation rate at twelve months rises to 10.29%, compared to 8.34% under HAR-QR. These two facts are not contradictory: they reflect the asymmetric check function and how the ENSO coefficient recalibrates the forecast boundary across climate regimes. At $\tau = 0.95$, the positive ENSO coefficient raises the boundary during El Niño episodes and lowers it when the Niño 3.4 anomaly is negative (La Niña). The lower boundary during La Niña conditions is breached more often when unexpected shocks occur, raising the overall violation rate. The quantile loss gain comes from El Niño episodes: when realized volatility spikes severely, the elevated boundary reduces the magnitude of exceedances substantially. The asymmetric check function penalizes large under-predictions more heavily than over-predictions, so the large-episode reductions outweigh the additional La Niña-period breaches in average loss. The net result is a model that is better calibrated *when it matters most* (large upside events) but less well calibrated overall.

Table 8: Coverage comparison: violation rates (%) for HAR-QR and HAR-X-QR

Horizon	Upper tail ($\tau = 0.95$)	
	HAR-QR	HAR-X-QR
1d	4.84	4.80
1w	5.15	5.22
1m	5.68	6.13
3m	5.91	6.49
6m	7.04	8.91
12m	8.34	10.29

Note: Under correct specification, the violation rate should equal 5%. Values closer to 5% indicate better calibration. $N \approx 10,400$ –10,500 forecast–realization pairs per horizon.

5.4 Statistical Significance: Model Confidence Sets

The preceding sections compare models by point estimates of loss functions. To assess whether the observed differences are statistically significant, we apply the Model Confidence Set (MCS) procedure of Hansen et al. (2011). The MCS identifies the set $\mathcal{M}^*$ of models whose forecasting ability cannot be distinguished from that of the best model at a given confidence level. We use the MCS procedure and configuration described in Section 3.4.

Table 9 reports MCS results for QLIKE loss. HAR-OLS is the sole member of $\mathcal{M}^*$ at the one-day and one-week horizons ($p = 1.00$), confirming that its mean forecast superiority over GARCH and historical volatility is statistically significant at short horizons. From one month onward, the HAR-QR and HAR-X-QR median forecasts enter $\mathcal{M}^*$, indicating that their mean forecast accuracy is statistically indistinguishable from HAR-OLS at longer horizons. GARCH(1,1) is excluded from $\mathcal{M}^*$ at every horizon; its inferior QLIKE performance is not attributable to sampling variation.

Table 9: Model Confidence Set: QLIKE loss (mean forecasts)

Model		Horizon					
		1d	1w	1m	3m	6m	12m
Historical Vol	p	0.01	0.04	0.07	0.10	0.10	0.06
	$\mathcal{M}^*$					✓	
GARCH(1,1)	p	0.01	0.00	0.00	0.00	0.00	0.01
	$\mathcal{M}^*$						
HAR-OLS	p	1.00	1.00	1.00	1.00	1.00	1.00
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓
HAR-QR (med.)	p	0.00	0.04	0.10	0.38	0.30	0.25
	$\mathcal{M}^*$			✓	✓	✓	✓
HAR-X-QR (med.)	p	0.00	0.04	0.10	0.31	0.30	0.25
	$\mathcal{M}^*$			✓	✓	✓	✓

Note: MCS p -values from Hansen et al. (2011); max- t statistic, 10,000 stationary bootstrap replications, 90% confidence level. ✓ indicates membership in $\mathcal{M}^*$. Bold: $p = 1.00$ (best model).

HAR-QR and HAR-X-QR evaluated on their median ($\tau = 0.50$) forecasts.

Table 10 presents MCS results for quantile loss at $\tau = 0.95$, the central evaluation criterion for distributional forecasting. Historical volatility and GARCH(1,1) are excluded from $\mathcal{M}^*$ at every horizon ($p < 0.05$ throughout), confirming that their inferior quantile loss is statistically significant and not a product of sampling variation. From one month onward, $\mathcal{M}^*$ contains HAR-OLS alongside the two direct quantile regression models, indicating that the pseudo-quantile approach is statistically indistinguishable from HAR-QR at medium and long horizons. At short horizons, however, HAR-OLS is excluded ($p = 0.04$ at one day, $p = 0.02$ at one week), confirming that explicit quantile

estimation provides a statistically significant advantage where volatility dynamics are most state-dependent.

Within the HAR family, a revealing pattern emerges. At the one-day horizon, HAR-QR is the sole member of $\mathcal{M}^*$ ($p = 1.00$), with HAR-X-QR narrowly excluded ($p = 0.05$). From one week onward, both models coexist in $\mathcal{M}^*$. The critical shift occurs at three months: HAR-X-QR takes $p = 1.00$ from HAR-QR, indicating that the ENSO-augmented model becomes the top-ranked specification. This reversal persists at six and twelve months ($p = 1.00$ for HAR-X-QR versus $p = 0.18$ and $p = 0.19$ for HAR-QR). Both models remain in $\mathcal{M}^*$ throughout, meaning the data do not permit a statistically significant ranking between them. The consistent shift of the top rank to HAR-X-QR from three months onward is descriptively suggestive, but any interpretation as evidence for a structural ENSO channel requires the robustness assessment in Section 5.5.

Table 10: Model Confidence Set: quantile loss at $\tau = 0.95$

Model		Horizon					
		1d	1w	1m	3m	6m	12m
Historical Vol	p	0.04	0.00	0.00	0.00	0.00	0.00
	$\mathcal{M}^*$						
GARCH(1,1)	p	0.00	0.00	0.00	0.00	0.00	0.00
	$\mathcal{M}^*$						
HAR-OLS	p	0.04	0.02	0.17	0.23	0.17	0.15
	$\mathcal{M}^*$			✓	✓	✓	✓
HAR-QR	p	1.00	1.00	1.00	0.50	0.18	0.20
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓
HAR-X-QR	p	0.05	0.19	0.37	1.00	1.00	1.00
	$\mathcal{M}^*$		✓	✓	✓	✓	✓

Note: Same MCS configuration as Table 9. ✓ indicates membership in $\mathcal{M}^*$ at the 90% confidence level. Bold: $p = 1.00$ (best model). HAR-OLS pseudo-quantile forecasts are included alongside the direct quantile models. The shift of $p = 1.00$ from HAR-QR to HAR-X-QR at three months reflects the onset of HAR-X-QR's ranking advantage in the evaluation sample.

5.5 Robustness of the ENSO Signal

The preceding results establish that HAR-X-QR outperforms HAR-QR over the full evaluation sample, particularly at the twelve-month upper tail. Before interpreting this as evidence of a structural ENSO–cocoa volatility channel, we apply the three robustness diagnostics outlined in Section 3: temporal stability, lag structure, and cross-commodity replication. Each addresses a distinct way in which the full-sample gain could be sample-specific rather than generalisable.

Temporal stability: expanding-window diagnostic

To assess whether the ENSO gain is concentrated in the 2022–2024 supply crisis, we trace the *cumulative* gain as the evaluation horizon advances through time. For each calendar month T in the out-of-sample period, define

$$\Delta\text{QL}(T) = \text{QL}_{\text{HAR-X-QR}}(\{t \leq T\}) - \text{QL}_{\text{HAR-QR}}(\{t \leq T\}),$$

where each loss is computed over all forecast–realisation pairs with evaluation date up to and including T . Negative values indicate that ENSO augmentation has been beneficial on a cumulative basis up to that point. The construction is fully recursive: no break date is assumed or imposed, and the curve spans the complete out-of-sample window without any ex-post partitioning.

Figure 7 shows that the full-sample ENSO gain is almost entirely a product of the 2022–2024 extreme-volatility episode, and that this concentration is specific to the upper tail.

At the twelve-month horizon and $\tau = 0.95$, the cumulative ΔQL remains positive (ENSO augmentation harmful on net) or indistinguishable from zero throughout the period prior to 2022. The curve then falls sharply during 2023 and 2024, with the entire full-sample improvement of 25.4% accumulating within those two years. At the six-month horizon the pattern is similar: a modest negative signal builds gradually from around 2014, but the dominant movement occurs in the 2023–2024 window, when the six-month curve falls from near zero to its full-sample endpoint of $\Delta\text{QL} = -0.67 \times 10^{-4}$.

The median ($\tau = 0.50$) and $\tau = 0.75$ columns tell a qualitatively different story. Across all four horizons those panels show irregular, oscillating paths that cross zero repeatedly and never accumulate a sustained downward trend, even during the 2022–2024 crisis window. The ENSO augmentation gain is therefore not a broad distributional shift but a recalibration concentrated in the extreme upper tail. This is consistent with the mechanism described in Section 5.3: the positive ENSO coefficient raises the $\tau = 0.95$ boundary during El Niño episodes, reducing the magnitude of large upside exceedances, while leaving the center of the distribution largely unaffected.

This result sharpens the robustness assessment. The expanding-window diagnostic shows *when* the gain first accumulates and *where* in the distribution it is concentrated. For the twelve-month upper tail, the answer to both questions is the same: almost exclusively during the 2022–2024 supply crisis. The reported 25.4% full-sample improvement reflects a gain that accrued in the final two years of the evaluation period and is absent from the median and $\tau = 0.75$ forecasts throughout.

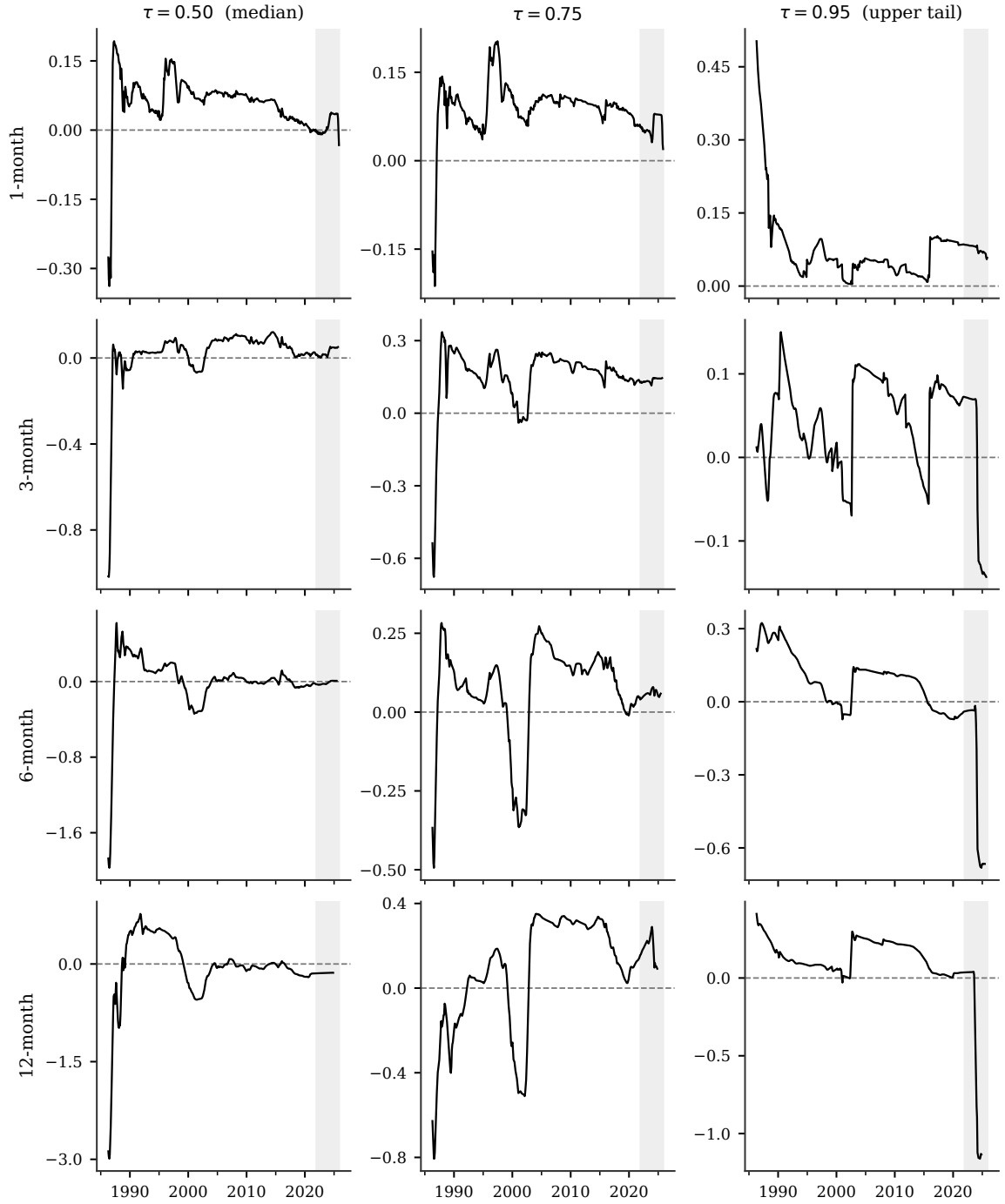


Figure 7: Expanding-window cumulative ENSO augmentation gain (ΔQL) across horizons and quantile levels

Notes: Each panel shows $\Delta QL(T)$ computed over all out-of-sample pairs with evaluation date $\leq T$. Negative values indicate cumulative net gain from ENSO augmentation. Rows: forecast horizons (1, 3, 6, 12 months). Columns: quantile levels $\tau \in \{0.50, 0.75, 0.95\}$. Shaded region marks the post-2022 supply-crisis episode for orientation; no formal cutoff is assumed. Minimum 500 observations before plotting.

Lag structure

If ENSO drives cocoa volatility through the agronomic-transmission chain (ENSO anomaly $\rightarrow$ West African rainfall $\rightarrow$ growing-season disruption $\rightarrow$ harvest shortfall $\rightarrow$ supply tightening $\rightarrow$ price volatility), the predictive power of the Niño 3.4 index should *strengthen* as the lag approaches 6–12 months, the time over which this chain unfolds. To test this, Table 11 reports quantile loss at $\tau = 0.95$ for HAR-X-QR with ENSO lagged 0, 1, 3, and 6 months.

Table 11: Lag-decay diagnostic: quantile loss ($\tau = 0.95$) by ENSO lag

Horizon	ENSO lag (months)			
	0	1	3	6
1m	0.000464	0.000461	0.000456	0.000457
6m	0.000387	0.000384	0.000381	0.000407
12m	0.000333	0.000321	0.000356	0.000416
HAR-QR baseline	0.000458	0.000454	0.000433	0.000446

Note: Lower values indicate better forecasts. Bold: best lag for each horizon. Lag k uses the Niño 3.4 value from k calendar months prior to the forecast date; construction is fully causal.

HAR-QR baseline is the corresponding no-ENSO model.

The results contradict the physical-channel prediction. At the twelve-month horizon, predictive power *peaks at lag 1* and decays sharply at lags 3 (0.000356) and 6 (0.000416), the latter performing nearly as poorly as the no-ENSO baseline (0.000446). At the six-month horizon, the pattern is similar: the best lag is 3, and lag 6 already deteriorates to 0.000407, well above the contemporaneous value. Under the agronomic-transmission hypothesis, lags 6 and 12 should be the most informative; the data show the opposite. The contemporaneous and near-contemporaneous specification outperforms all longer lags, which is the pattern expected of a regime-state indicator rather than a causal leading predictor.

This is further confirmed by the cross-correlogram in Figure 8, which plots the Pearson correlation between monthly Niño 3.4 and monthly mean cocoa volatility at lags -24 to $+24$ months. Positive lags indicate that ENSO leads volatility. In the full sample, the correlation is statistically significant only at lags 0, 1, and 2 on the positive side; the physical-channel prediction window (lags $+6$ to $+12$) is entirely insignificant. In the subsample excluding the supply-crisis period, correlations are significant across a wider range of lags, but this reflects the strong autocorrelation of the Niño 3.4 index itself: once ENSO is negative, it tends to remain so for 6–12 months, inducing apparent significance at distant lags through ENSO’s own persistence rather than through independent predictive relationships.

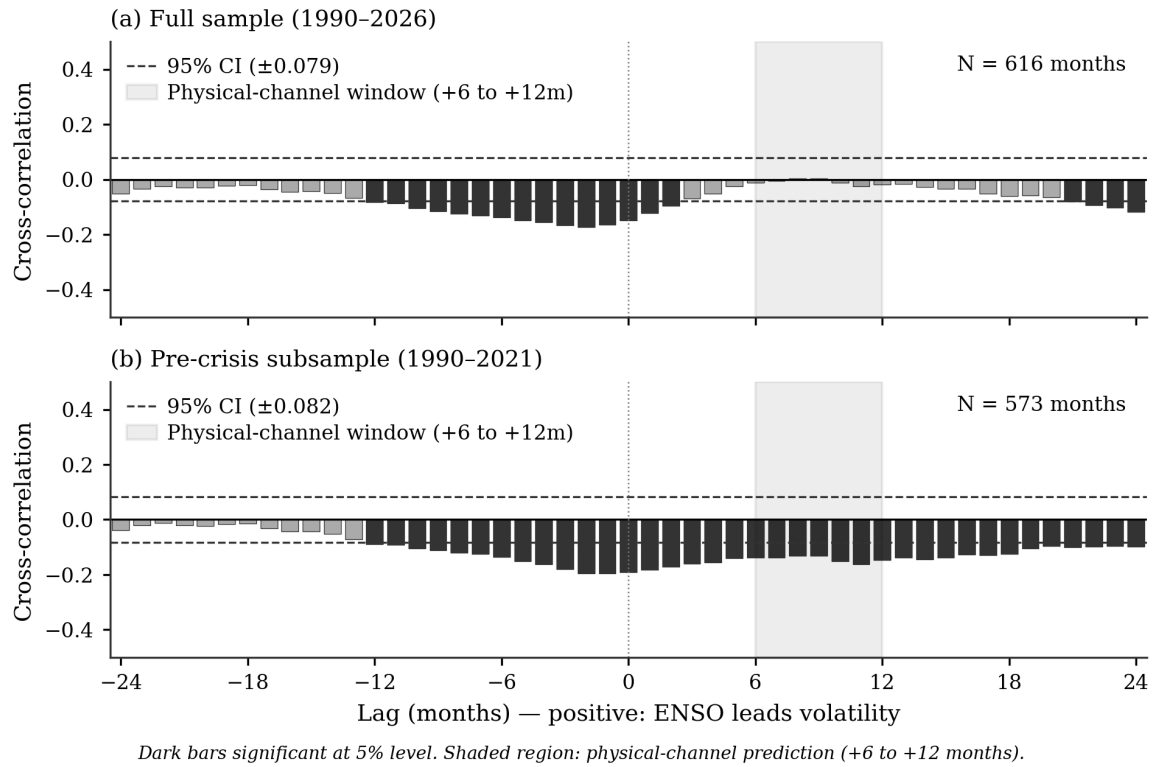


Figure 8: Cross-correlogram: Niño 3.4 vs. monthly mean cocoa OHLC volatility

Notes: Pearson correlation at lags -24 to $+24$ months. Positive lag k : Niño 3.4 at time t correlated with volatility at $t + k$ (ENSO leads). Dark bars exceed the 95% confidence interval ($\pm 1.96/\sqrt{N}$).

Shaded region: physical-channel prediction window (+6 to +12 months). Panel (a): full sample.

Panel (b): subsample excluding observations from 2022 onward (supply-crisis period).

Cross-commodity validation

If ENSO operates as a genuine climate channel it should produce comparable predictive gains on other tropical commodities exposed to the same Pacific climate signal. Arabica coffee and raw sugar are natural comparators: both are weather-sensitive soft commodities that share downstream markets and exhibit volatility spillovers with cocoa (Ge et al. 2025). Table 12 reports the $\tau = 0.95$ quantile loss improvement from ENSO augmentation for Arabica coffee (ICE KC=F, USD/lb) and raw sugar (ICE SB=F, USD cents/lb), estimated using the identical HAR-X-QR specification and rolling-window setup.

Table 12: Cross-commodity ENSO gain (ΔQL , $\tau = 0.95$) at 6- and 12-month horizons

Horizon	Cocoa	Coffee	Sugar
6m	−0.000067	+0.000014	−0.000014
12m	−0.000114	+0.000031	−0.000005

Note: $\Delta QL = QL(\text{HAR-X-QR}) - QL(\text{HAR-QR})$. Negative: ENSO augmentation improves forecasts. Arabica coffee: ICE KC=F continuous front-month, $N \approx 3,870$ out-of-sample pairs. Raw sugar: ICE SB=F continuous front-month, $N \approx 3,832$ out-of-sample pairs. Same Yang-Zhang volatility estimator and rolling-window setup as cocoa.

ENSO augmentation does not replicate on either comparator. For coffee, ENSO *worsens* long-horizon upper-tail forecasts at both horizons, with a +0.000031 deterioration at twelve months. For sugar, the gain is −0.000005 at twelve months, a factor of 22 smaller than cocoa’s −0.000114. A structural climate channel would predict qualitatively similar gains across all three tropical commodities grown in the same equatorial belt; the data show a pattern that is specific to cocoa and its sample period.

Summary assessment of RQ2 robustness

The diagnostics present a consistent picture. The ENSO augmentation gain over the full evaluation sample is numerically substantial, and HAR-X-QR holds the top MCS rank from three months onward, but its temporal stability is limited. The expanding-window diagnostic (Figure 7) shows that cumulative ENSO augmentation provided no net benefit at the twelve-month $\tau = 0.95$ horizon throughout the three decades preceding the 2022–2024 supply crisis; the entire 25.4% full-sample improvement accumulates within the 2023–2024 window. At the six-month horizon the pattern is similar. Crucially, the matrix figure also shows that the gain is absent from the median and $\tau = 0.75$ forecasts throughout the evaluation period, confirming that the ENSO effect is a tail-specific, crisis-concentrated phenomenon rather than a broad improvement in distributional accuracy. The remaining diagnostics reinforce this reading: (i) the lag structure is inconsistent with the agronomic-transmission hypothesis — predictive

power peaks at near-contemporaneous lags rather than at the 6–12 month lags the physical channel predicts, (ii) cross-commodity validation does not replicate the result, and (iii) replacing the continuous Niño 3.4 index with rolling-standard-deviation variants (12- and 36-month windows) produces the same crisis-concentrated pattern, confirming that the result is not specific to the index operationalisation. The answer to RQ2 is therefore that ENSO augmentation produces a numerically large improvement in upper-tail forecasts at long horizons over the full evaluation sample, but this improvement is driven by the 2022–2024 extreme-volatility episode rather than by a persistent predictive relationship. The gain is best described as a sample-period-specific correlation between the ENSO state and the cocoa supply crisis, rather than evidence of a replicable structural climate channel.

5.6 Summary of Findings

Table 13 consolidates the key metrics for all models across the horizon spectrum. Three conclusions emerge.

Table 13: Consolidated model comparison across horizons

Metric	Model	Horizon				
		1d	1m	3m	6m	12m
QLIKE	Historical Vol	−7.002	−7.327	−7.334*	−7.324	−7.310
	GARCH(1,1)	−7.242	−7.406*	−7.381	−7.356	−7.345
	HAR-OLS	−7.302	−7.521*	−7.515	−7.492	−7.485
QL ($\tau=0.95$)	Historical Vol	0.001502	0.000871*	0.000889	0.000914	0.000930
	GARCH(1,1)	0.001393	0.001011*	0.001053	0.001066	0.001044
	HAR-OLS	0.001272	0.000507	0.000472	0.000479	0.000451*
	HAR-QR	0.001193	0.000458	0.000433*	0.000454	0.000446
	HAR-X-QR	0.001195	0.000464	0.000418	0.000387	0.000333*

Note: Bold: best model at each horizon. *Best horizon for each model. QLIKE: more negative is better. QL: lower is better. The 1w horizon is omitted for compactness.

First, regarding RQ1, HAR-OLS achieves the lowest QLIKE at all horizons; HAR-QR achieves the lowest upper-tail quantile loss at horizons up to one month and remains competitive at three months (3.6% above the best model). At six and twelve months the ENSO-augmented variant takes the lead, with HAR-QR falling 17% and 34% behind HAR-X-QR respectively. Direct quantile estimation outperforms the HAR-OLS pseudo-quantile at every horizon, with the advantage narrowing from 10% at one month to less than 2% at twelve months. The same QLIKE ranking holds under MSE and MAE (Figure 3), confirming that the result is not an artefact of the chosen loss function. The MCS confirms statistical significance: GARCH(1,1) is excluded from $\mathcal{M}^*$ at every horizon for both loss criteria (Tables 9 and 10).

Second, regarding RQ2, the Niño 3.4 index produces a substantial improvement at long horizons over the full evaluation sample: the $\tau = 0.95$ quantile loss falls by 25.44% at twelve months (Table 7), and HAR-X-QR holds the top MCS rank from three months onward (Table 10). However, robustness testing in Section 5.5 shows that this gain does not survive temporal disaggregation: it vanishes entirely in the pre-2022 subsample, is not replicated on comparable tropical commodities, and is inconsistent with the lag structure the agronomic-transmission hypothesis predicts. The result is best described as a sample-specific finding rather than evidence of a structural ENSO–cocoa volatility channel.

Third, coverage analysis reveals a systematic challenge at long horizons. All models produce violation rates above the nominal 5% at six and twelve months, indicating that the conditional distribution of cocoa volatility is harder to characterize at procurement-relevant horizons than at short ones. ENSO augmentation introduces a recalibration effect at the upper tail. These limitations are discussed further in Section ??.

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Appendices

A Supplementary Tables

This appendix collects reference material and detailed estimates that support the main text but are not essential to the line of argument: the data description (Table 14), the unit root tests (Table 15), the sup-Wald structural break test (Table 16), the validation of the volatility proxy against intraday realized volatility (Table 17), and the full coefficient estimates of the HAR-QR and HAR-X-QR models (Table 18).

Table 14: Summary of Cocoa Futures Data

Characteristic	Value
Exchange	ICE Futures Europe
Contract	LCC (Cocoa Futures)
Contract size	10 metric tonnes
Price quotation	GBP per metric tonne
Continuous series	LCCc1 (front-month rolling)
Data source	Refinitiv Workspace
Sample period (daily)	April 1, 1974 – December 15, 2025
Number of observations (daily)	13,090 trading days
Sample period (5-minute)	February 3, 2025 – January 12, 2026
Number of observations (5-minute)	20,791 intraday observations
Variables (daily)	Open, High, Low, Close, Volume
Variables (5-minute)	Open, High, Low, Close, Volume, Bid, Ask

Table 15: Unit root tests for stationarity

Series	<i>N</i>	ADF		PP	
		Statistic	<i>p</i> -value	Statistic	<i>p</i> -value
Log returns	13,087	−112.08	< 0.001	−112.09	< 0.001
Yang-Zhang volatility	13,087	−8.42	< 0.001	−90.99	< 0.001
Niño 3.4 anomaly	787	−6.85	< 0.001	−6.46	< 0.001

Note: ADF = Augmented Dickey–Fuller test (Dickey and Fuller 1979); PP = Phillips–Perron test (Phillips and Perron 1988). Both tests include a constant. ADF lag length selected by AIC. The null hypothesis is a unit root; rejection indicates stationarity.

Table 16: Sup-Wald structural break test for the volatility series

Quantity	Value
Series tested	Yang-Zhang volatility (out-of-sample)
Candidate window	1 Jan 2022 – 1 Jan 2025
Estimated break date	8 February 2024
Sup-Wald F -statistic	1,809.05
Critical value (10%)	5.74
Critical value (5%)	7.17
Critical value (1%)	10.17
Pre-break observations	10,098
Post-break observations	470
Mean volatility, pre-break	0.0134
Mean volatility, post-break	0.0313
Post/pre ratio	2.34×

Note: Single-break sup-Wald test (Andrews 1993) applied to the out-of-sample Yang-Zhang volatility series. The test fits a mean-shift regression at each candidate date and reports the maximum F -statistic; the null hypothesis of no break is rejected at the 1% level. Critical values from Andrews (1993), Table 1 ($k = 1$, mean shift).

Table 17: Validation of OHLC volatility estimators against intraday realized volatility

Estimator	Pearson correlation with RV		
	Daily	Weekly	Monthly
Yang-Zhang	0.681	0.773	0.921
Garman-Klass	0.697	0.796	0.927
Parkinson	0.675	0.796	0.933
Close-to-close	0.383	0.635	0.880

Note: Pearson correlation between each daily OHLC-based volatility estimator and realized volatility (RV) computed from five-minute intraday returns, over the period 3 February 2025 – 12 January 2026 (196 trading days). The weekly and monthly columns aggregate both series to 5- and 22-day averages before computing the correlation. The rising correlation with aggregation reflects the averaging-out of daily estimation noise; the Yang-Zhang estimator tracks RV closely at the monthly frequency relevant to the forecasting exercise.

Table 18: Full-sample HAR-QR and HAR-X-QR coefficient estimates

Horizon	Model	β_0	β_d	β_w	β_m	β_{ENSO}
<i>Panel: $\tau = 0.50$</i>						
1 day	HAR-QR	0.0022	0.1035	0.1837	0.4375	—
	HAR-X-QR	0.0023	0.1030	0.1801	0.4345	-0.0001
1 week	HAR-QR	0.0027	0.0670	0.1712	0.5182	—
	HAR-X-QR	0.0028	0.0659	0.1729	0.5127	-0.0001
1 month	HAR-QR	0.0040	0.0335	0.1332	0.5260	—
	HAR-X-QR	0.0041	0.0335	0.1299	0.5224	-0.0002
3 months	HAR-QR	0.0048	0.0214	0.0657	0.5553	—
	HAR-X-QR	0.0049	0.0208	0.0682	0.5437	-0.0003
6 months	HAR-QR	0.0061	0.0174	0.0420	0.5096	—
	HAR-X-QR	0.0061	0.0164	0.0464	0.5017	-0.0002
12 months	HAR-QR	0.0073	0.0098	0.0282	0.4446	—
	HAR-X-QR	0.0073	0.0099	0.0270	0.4438	-0.0001
<i>Panel: $\tau = 0.75$</i>						
1 day	HAR-QR	0.0024	0.1508	0.3087	0.5787	—
	HAR-X-QR	0.0026	0.1500	0.3096	0.5660	-0.0001
1 week	HAR-QR	0.0029	0.0849	0.2391	0.6316	—
	HAR-X-QR	0.0029	0.0851	0.2377	0.6313	-0.0000
1 month	HAR-QR	0.0040	0.0453	0.1637	0.6573	—
	HAR-X-QR	0.0041	0.0480	0.1527	0.6574	-0.0002
3 months	HAR-QR	0.0052	0.0278	0.0576	0.6845	—
	HAR-X-QR	0.0052	0.0298	0.0578	0.6794	-0.0002
6 months	HAR-QR	0.0064	0.0184	0.0460	0.6276	—
	HAR-X-QR	0.0065	0.0178	0.0460	0.6191	-0.0002
12 months	HAR-QR	0.0079	0.0123	0.0377	0.5368	—
	HAR-X-QR	0.0079	0.0121	0.0382	0.5371	0.0000
<i>Panel: $\tau = 0.95$</i>						
1 day	HAR-QR	0.0030	0.2859	0.4661	1.0682	—
	HAR-X-QR	0.0032	0.2908	0.4583	1.0570	-0.0001
1 week	HAR-QR	0.0032	0.1796	0.3431	0.8998	—
	HAR-X-QR	0.0031	0.1754	0.3400	0.9178	-0.0004
1 month	HAR-QR	0.0034	0.0517	0.2189	1.0066	—
	HAR-X-QR	0.0034	0.0439	0.2456	0.9896	-0.0003
3 months	HAR-QR	0.0052	0.0255	0.1667	0.9463	—
	HAR-X-QR	0.0052	0.0252	0.1649	0.9494	-0.0000
6 months	HAR-QR	0.0068	0.0285	0.0923	0.8527	—
	HAR-X-QR	0.0073	0.0321	0.0905	0.8275	0.0004
12 months	HAR-QR	0.0084	0.0235	0.1339	0.7131	—
	HAR-X-QR	0.0098	0.0334	0.1073	0.6982	0.0014

Note: Quantile regression coefficients estimated once on the full sample for each horizon and quantile τ . β_0 is the intercept; β_d , β_w , β_m are the daily, weekly and monthly HAR components; β_{ENSO} is the Niño 3.4 loading (HAR-X-QR only). The table is descriptive: it reports the sign and magnitude of the loadings only. Significance tests are omitted because the forecast targets are overlapping h -day-ahead averages, which violates the independence assumption underlying the standard errors. The forecasting results in Section 5 use rolling-window re-estimation; these full-sample estimates are reported for reference only.